

Journal of Speech, Language, and Hearing Research

Synthetic Data in Communication Sciences and Disorders Promoting an Open, Reproducible, and Cumulative Science

--Manuscript Draft--

Manuscript Number:	
Full Title:	Synthetic Data in Communication Sciences and Disorders Promoting an Open, Reproducible, and Cumulative Science
Article Type:	Research Article
Section/Category:	Speech
Funding Information:	
Keywords:	Open data; Reproducibility; Meta-science
Manuscript Classifications:	Research Issues and Methodology; Open Science; Rigor and Transparency
Abstract:	Purpose Reproducibility is a core principle of science and access to a study's data is essential to reproduce its findings. However, data sharing is uncommon in the field of Communication Sciences and Disorders (CSD), often due to concerns related to privacy and disclosure risks. Synthetic data offers a potential solution to this barrier by generating artificial datasets that do not represent real individuals yet retain statistical properties and relationships from the original data. This study evaluates the performance of synthetic data generation using open data from previously published studies across the American Speech-Language-Hearing Association (ASHA) 'Big Nine' domains. Method Open datasets were obtained from previously published research within the ASHA domains of articulation, cognition, communication, fluency, hearing, language, social communication, voice and resonance, and swallowing. Synthetic datasets were generated with the synthpop R package. Inferential statistics (p-values) and effect sizes from synthetic datasets were compared to those from the original datasets. Results Synthetic datasets maintained the direction of p-values in six out of the nine studies and effect size categorizations in five studies. In cases where synthetic datasets did not maintain 95% of the inferential or effect size results, the absolute mean difference between synthetic and original datasets was relatively low, suggesting that the distribution of results from synthetic datasets closely approximated the alpha or effect size categorization threshold. Conclusion Findings suggest that synthetic data can effectively maintain statistical properties and relationships across a wide range of data commonly seen in the field of CSD. While some studies with fewer observations than recommended (i.e., $n < 130$) showed lower agreement and greater variability in p-values and effect size estimates, this was not consistently appreciated. Therefore, researchers who use synthetic data should assess its stability in preserving their results. This study concludes with a general framework on sharing open data to facilitate computational reproducibility and foster a cumulative science in the field of CSD.
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Synthetic Data in Communication Sciences and Disorders

Synthetic Data in Communication Sciences and Disorders: Promoting an Open, Reproducible, and Cumulative Science

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Disclosures:

The authors have no financial or non-financial disclosures.

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Authorship Contributions (CRediT taxonomy - <https://casrai.org/credit/>)

Author Roles: ¹conceptualization, ²data curation, ³formal analysis, ⁴funding acquisition, ⁵investigation, ⁶methodology, ⁷project administration, ⁸resources, ⁹software, ¹⁰supervision,

22 ¹¹validation, ¹²visualization, ¹³writing – original draft, ¹⁴writing – reviewing & editing

23 JCB: 1, 2, 3, 6, 9, 12, 13

24 AT: 1, 2, 3, 6, 9, 11, 12, 14

25 EK: 1, 2, 3, 6, 9, 11, 12, 14

26

27 **Funding:** None.

28

29 **Ethical Approval:** This study was deemed exempt by the Human Research Ethics Committee at
30 the University of Queensland (#2024/HE001484).

31

32 **Keywords:** Open data; Reproducibility; Meta-science; Communication sciences and disorders

33

34 **Study Preregistration and Data Availability:** The study preregistration (<https://osf.io/vhggq2>)
35 and associated data and analysis scripts (<https://osf.io/yhkqf/>) are publicly available on the Open
36 Science Framework.

37

38 **Abstract**

39 **Purpose:** Reproducibility is a core principle of science and access to a study's data is essential to
40 reproduce its findings. However, data sharing is uncommon in the field of Communication
41 Sciences and Disorders (CSD), often due to concerns related to privacy and disclosure risks.
42 Synthetic data offers a potential solution to this barrier by generating artificial datasets that do
43 not represent real individuals yet retain statistical properties and relationships from the original
44 data. This study evaluates the performance of synthetic data generation using open data from
45 previously published studies across the American Speech-Language-Hearing Association
46 (ASHA) 'Big Nine' domains.

47 **Method:** Open datasets were obtained from previously published research within the ASHA
48 domains of articulation, cognition, communication, fluency, hearing, language, social
49 communication, voice and resonance, and swallowing. Synthetic datasets were generated with
50 the *synthpop* R package. Inferential statistics (*p*-values) and effect sizes from synthetic datasets
51 were compared to those from the original datasets.

52 **Results:** Synthetic datasets maintained the direction of *p*-values in six out of the nine studies and
53 effect size categorizations in five studies. In cases where synthetic datasets did not maintain 95%
54 of the inferential or effect size results, the absolute mean difference between synthetic and
55 original datasets was relatively low, suggesting that the distribution of results from synthetic
56 datasets closely approximated the alpha or effect size categorization threshold.

57 **Conclusion:** Findings suggest that synthetic data can effectively maintain statistical properties
58 and relationships across a wide range of data commonly seen in the field of CSD. While some
59 studies with fewer observations than recommended (i.e., $n < 130$) showed lower agreement and
60 greater variability in *p*-values and effect size estimates, this was not consistently appreciated.

61 Therefore, researchers who use synthetic data should assess its stability in preserving their
62 results. This study concludes with a general framework on sharing open data to facilitate
63 computational reproducibility and foster a cumulative science in the field of CSD.

64

65 **Introduction**

66 Transparency and openness are fundamental tenets of science, with computational
67 reproducibility playing a key role in maintaining these values. Computational reproducibility
68 refers to the ability to recreate a study's results using the original data. Nowadays, the vast
69 majority of scientific studies use some degree of computation in processing data, conducting
70 descriptive or inferential statistics, and visualizing results. When these computations are
71 reproducible, the transparency and confidence in findings are enhanced. Achieving
72 computational reproducibility, however, requires authors to share their data. Both the National
73 Institutes of Health and the National Science Foundation mandate data sharing and management
74 plans to ensure that scientific data supporting a study is shared upon publication and aligns with
75 FAIR (Findability, Accessibility, Interoperability, and Reuse) principles of digital assets (Watson
76 et al., 2023; Wilkinson et al., 2016).

77 Providing open, publicly available data benefits scientists, funding bodies, and society at
78 large by enabling researchers to verify results, generate new knowledge (e.g., meta-analyses,
79 secondary analyses), develop hypotheses, and minimize redundant data collection (Chow et al.,
80 2023). In this sense, sharing data promotes a cumulative and self-correcting science. Despite the
81 clear benefits of open data and its growing adoption in other fields like psychology and the
82 biobehavioral sciences (Quintana, 2020), only 26% of a sample of researchers in the field of
83 Communication Sciences and Disorders (CSD) reported sharing their data publicly at least once
84 (El Amin et al., 2023).

85 Understanding the nuances of data sharing requires a closer look at the different types of
86 data generated throughout a research project's life cycle. These include raw collected data,
87 processed intermediate data, and final analysis data (Table 1). However, a common

misconception is that open data refers solely to sharing raw data (e.g., audio recordings, videos, MRI data) (Pfeiffer et al., 2024). In reality, sharing intermediate or analysis data can also support reproducibility while reducing privacy and confidentiality concerns associated with sharing raw data. However, these different types of data offer varying levels of utility: sharing raw data enables maximum reproducibility and secondary research opportunities, while analysis data (although easier to share) primarily supports computational reproducibility.

Table 1 here.

Both individual and system-level barriers hinder data sharing, including a lack of time, knowledge, support from colleagues, and perceived incentives (Pfeiffer et al., 2024). Furthermore, each type of data comes with unique challenges regarding data sharing. For raw data, it is common that researchers often do not obtain consent to share data or cannot contact participants after data collection. Additionally, sharing de-identified raw or intermediate data may require additional approval from the institutional review board. Even when de-identification is possible, anonymized intermediate or analysis datasets can still carry re-identification risks, especially in small samples or vulnerable populations where indirect identifiers (e.g., gender, age, or race) may compromise participant confidentiality (Rocher et al., 2019). Therefore, although sharing de-identified analysis data is the minimum requirement for ensuring computational reproducibility and promoting cumulative science, concerns about privacy and confidentiality may persist.

Synthetic data generation offers a potential solution to maintaining participants' privacy and confidentiality in publicly available datasets (Drechsler & Haensch, 2024; Rubin, 1993). Synthetic data involves creating an artificial dataset that does not represent real individuals, ensuring no risk of disclosure since participants in the synthetic dataset do not correspond to real

111 individuals. Importantly, synthetic data retains the statistical properties and relationships of the
112 original data, allowing researchers to reproduce study findings, explore the dataset, and develop
113 new questions and hypotheses. Synthetic data generation is widely used across medical research,
114 industry, and government agencies, most notably by the United States Census Bureau (Jarmin et
115 al., 2014). Though synthetic data methods were first proposed more than 30 years ago (Rubin,
116 1993), recent analytic and software developments have made it easier and more efficient to
117 generate high-quality synthetic data (Nowok et al., 2016).

118 Despite the potential utility of synthetic data to promote data sharing in the field of CSD,
119 this approach is not widely known or adopted in the field. Data commonly collected in CSD
120 research poses unique challenges, including smaller sample sizes than are typically
121 recommended for synthetic data generation and a wide range of outcomes and analyses (Borders
122 et al., 2022; Gaeta & Brydges, 2020). Therefore, the present study aimed to examine the utility
123 of synthetic data generation with open datasets from the ‘Big Nine’ American Speech-Language-
124 Hearing Association (ASHA) domains. We hypothesized that synthetic datasets would maintain
125 the statistical properties and relationships (i.e., p -value and effect size) of the original datasets,
126 and that synthetic data would remain stable when generating multiple datasets. A secondary goal
127 was to provide a framework to describe considerations when sharing data, thereby addressing
128 concerns regarding researcher knowledge and participant confidentiality in open data.

129 **Method**

130 *Description of Original Datasets from ASHA ‘Big Nine’ Domains*

131 Authors performed a manual search to obtain publicly available datasets from previously
132 published research articles related to the ‘Big Nine’ ASHA domains: swallowing (Curtis et al.,

2023), articulation (Thompson et al., 2023), fluency (Elsherif et al., 2021), voice and resonance (Novotný et al., 2016), hearing (Battal et al., 2019), communication modalities (King et al., 2022), receptive and expressive language (Kearney et al., 2023), cognitive aspects of communication (Clough et al., 2023), and social aspects of communication (Chanchaochai & Schwarz, 2023). Authors then reproduced an analysis from each study. Table 2 provides a description of the population, analysis, and open materials for each study.

Table 2 here.

Generation of Synthetic Datasets and Comparison with Original Dataset

Synthetic data generation and statistical analyses were conducted in R version 4.2.1 (R Core Team, 2022). Synthetic data was generated with the *synthpop* R package (version 1.8.0) (Nowok et al., 2016). Specifically, *synthpop* uses a non-parametric classification and regression tree approach that generates data by sampling from a probability distribution and can handle any type of data. Our aims were twofold: (1) to determine whether a synthetic dataset maintained the statistical properties and relationships of the original dataset and (2) to examine whether this remained stable when generating multiple synthetic datasets. In light of these aims, our approach involved generating 100 different synthetic datasets for each original dataset from an ASHA ‘Big Nine’ domain. A statistical model with the original dataset was fit, and the p -value and effect size were recorded. If 95% of p -values and effect sizes from the synthetic datasets demonstrated a similar result as the original study, then this indicated that synthetic data maintained the statistical relationship. Specifically, we further defined this as a similar inferential result for p -values (i.e., a ‘significant’ or ‘non-significant’ p -value based on the original study’s alpha level) and effect sizes that maintained their categorization based on conventional thresholds (e.g., a

‘medium’ effect size). Measures of effect size and their interpretation for each study are provided in Table 3. If variability between the 100 synthetic datasets was appreciated, we visualized and described the dispersion of this distribution. The analysis plan for this study was preregistered on the Open Science Framework (<https://osf.io/vhggq2>).

[Table 3 here.](#)

In addition to these inferential comparisons, we provide a brief tutorial to guide the reader through the required steps to generate synthetic data. This is accomplished in the context of two datasets (Curtis et al., 2023; Thompson et al., 2023) with additional data visualization and detailed R code. Since Curtis et al. (2023) did not perform inferential tests, we directly compared each synthetic dataset to the original data with a zero-inflated beta multilevel model with the *gamlss* package (version 5.4.3) (Stasinopoulos & Rigby, 2007). This model included fixed effects of dataset type (synthetic/original) and bolus consistency (thin liquid/extremely thick/regular), and a random intercept of participant. Due to issues with model convergence, the fixed effect structure was simplified to only include dataset type. The p -value from both zero-inflated and beta portions of the model were evaluated and $p < .05$ was interpreted no statistically significant difference between the synthetic and original dataset.

Results

The tutorial data and accompanying code can be accessed on the Open Science Framework (<https://osf.io/yhkqf/>). To get started, download R (<https://cran.r-project.org/>) and an interface like RStudio (<https://posit.co/download/rstudio-desktop/>). Open the *open-and-synthetic-data.Rproj* file in RStudio. To reproduce the tutorial code only, open the file called *tutorial_script.R*. To reproduce the manuscript, open the file called *synthetic_manuscript.qmd*.

Study 1: Normative Reference Values for Swallowing Outcomes

Curtis et al. (2023) examined normative reference values for swallowing outcomes during flexible endoscopic evaluations of swallowing among 39 non-dysphagic, community-dwelling adults. In this observational cohort study, participants were administered 15 swallowing trials that varied by bolus size, consistency, contrast agent, and swallowing instructions. A variety of swallowing outcomes were measured, including the amount of laryngeal vestibule residue rated with the Visual Analysis of Swallowing Efficiency and Safety. Median and interquartile ranges (IQR) were used to describe the distribution of laryngeal vestibule residue ratings.

To generate synthetic data, we first load in the original dataset, wrangle the dataset using the *tidyverse* collection of packages (Wickham et al., 2019) (v. 1.3.2). The data wrangling steps include (1) loading required R packages, (2) importing the original dataset csv file, (3) reformatting variable names for consistency and readability, (4) selecting the variables needed for the analysis, (5) converting appropriate categorical variables to factors, and (6) calculating the laryngeal vestibule severity rating as a percentage. These steps are shown below with the following commands:

```
# Load required packages
library(tidyverse) # data wrangling
library(synthpop) # R package to generate synthetic data

# Load original data
swallowing_original_data <-
  # read csv file from appropriate path
  read.csv(here::here("Data/01_Swallowing/norms_ratings.csv")) |>
  # clean variable names
  janitor::clean_names() |>
  # select only relevant variables from dataset
  dplyr::select(c(study_id, bolus_consistency,
                  laryngeal_vestibule_severity_rating)) |>
  mutate(
    # convert study_id and bolus_consistency to factors
    study_id = as.factor(study_id),
    bolus_consistency = as.factor(bolus_consistency),
    # express laryngeal_vestibule_severity_rating as a %
```

```

210     laryngeal_vestibule_severity_rating = laryngeal_vestibule_severity_rating/100
211   )

```

212 Next, we create a synthetic dataset with the `syn()` function from the *synthpop* package.

213 Within the function, ‘method’ specifies the synthesising method for the data. The default in

214 *synthpop* is “cart” (Classification and Regression Tree). If a synthetic dataset fails to generate

215 with this method, Nowok et al. (2018) recommend an alternative implementation of the CART

216 technique from package *party* (Hothorn et al., 2006). This dataset, for example, required the

217 ‘ctree’ CART specification. Next, specify the number of synthetic datasets to generate within

218 ‘m’. Once the synthetic dataset is generated, extract and convert it to a dataframe for additional

219 data wrangling and visualization.

```

220   # Create a synthetic dataset
221   synthetic_data <-
222     syn(swallowing_original_data, # name of the original data
223        method = "ctree", # CART model to generate synthetic data
224        m = 1 # number of synthetic datasets to generate
225     )
226
227   Synthesis
228   -----
229   study_id bolus_consistency laryngeal_vestibule_severity_rating
230
231   # Extract the synthetic dataset and convert into a data frame
232   synthetic_dataset <- as.data.frame(synthetic_data$syn)

```

232 An important step in this process is to assess the general utility of the synthetic dataset by

233 visualizing any obvious differences compared to the original dataset. This can be easily

234 accomplished with the `compare()` function in the *synthpop* package or manually with data

235 wrangling and the *ggplot* package. Figure 1 suggests that the synthetic dataset demonstrated

236 similar distributions for the variables of bolus consistency and laryngeal vestibule residue rating.

```

237   # Comparison of original and synthetic datasets with synthpop package
238   swallowing_comparison <- compare(
239     synthetic_data, # synthetic dataset
240     swallowing_original_data, # original dataset
241     vars = c("bolus_consistency",
242             "laryngeal_vestibule_severity_rating"), # variables for comparison

```

```

243 stat = "counts", # Present the raw counts for each variable
244 cols = c("#62B6CB", "#1B4965") # Setting the colours in the plot
245 )

```

Figure 1 here.

Descriptively, the synthetic dataset classified 64% of laryngeal vestibule ratings on thin liquid boluses as ‘absent’ (i.e., 0% residue) compared to 68% in the original dataset. In the synthetic dataset, the median value on thin liquids was 0.03 (IQR: 0.02 - 0.045) compared to 0.03 (IQR: 0.02 - 0.04) in the original dataset. 98.61% of extremely thick liquids were classified as having no laryngeal vestibule residue compared to 100% in the original dataset. A similar pattern was appreciated for regular solids (96.43% in synthetic vs. 100% in original dataset). When examined across 100 synthetic datasets, findings from the zero-inflated beta multilevel models indicate that 100% and 98% of synthetic datasets were not statistically significantly different than the original dataset for the zero-inflated and beta portions of the model, respectively (Table 4). Additionally, effect size categorizations were maintained for 100% of both zero-inflated and beta portions of the model.

Study 2: Vowel Acoustics as Predictors of Speech Intelligibility in Dysarthria

Thompson et al. (2023) examined the relationship between vowel space area and speech intelligibility among 40 speakers with dysarthria of varying etiologies, including Parkinson’s disease, amyotrophic lateral sclerosis, Huntington’s disease, and cerebellar ataxia. A linear regression model revealed a statistically significant relationship between vowel space area and intelligibility ($p < .001$) with a Cohen’s f of 0.59, corresponding to a conventionally “large” effect size (Table 3).

Below we import the original dataset, wrangle the data, and generate a synthetic data set. The data wrangling steps include (1) importing the original dataset, (2) removing the reliability

trials from the dataset, (3) removing a string character from the SpeakerID variable, and (4) selecting only the variables needed for the analysis.

```
# import original data
articulation_original_data <- rio::import(
  file = here::here("Data", "02_Articulation", "data_Acoustic Measures.csv")
) |>
# Remove the reliability trials that have "_rel" in the SpeakerID variable
dplyr::filter(
  !grepl(
    pattern = "_rel",
    x = SpeakerID
  )
) |>
# Selecting just the variables we need
dplyr::select(
  SpeakerID, # ID
  VSA_b, # VSA in Bark
  Int = Int_OT # intelligibility (orthographic transcriptions)
)
```

Next, we generate a synthetic dataset with the `syn()` function, extract the dataset, and convert it to a dataframe. *Synthpop* provides a warning message that this dataset has fewer observations than recommended (> 130).

```
# generate synthetic dataset
articulation_synthetic_dataset <- syn(articulation_original_data,
  m = 1,
  seed = 2024)

CAUTION: Your data set has fewer observations (40) than we advise.
We suggest that there should be at least 130 observations
(100 + 10 * no. of variables used in modelling the data).
Please check your synthetic data carefully with functions
compare(), utility.tab(), and utility.gen().

Variable(s): SpeakerID have been changed for synthesis from character to factor.

Synthesis
-----
SpeakerID VSA_b Int

# Extract the synthetic dataset and convert into a data frame
articulation_synthetic_dataset <- as.data.frame(articulation_synthetic_dataset$syn)
```

Next, we compare the distributions for vowel space area and speech intelligibility

between the synthetic and original dataset. Figure 2 suggests that while the synthetic data largely approximates the original dataset, there are several values that are oversampled in the synthetic dataset.

```
# Comparison of original and synthetic datasets with synthpop package
articulation_comparison <- compare(
  articulation_synthetic_dataset, # synthetic dataset
  articulation_original_data, # original dataset
  vars = c("VSA_b",
           "Int"), # variables for comparison
  stat = "counts", # Present the raw counts for each variable
  cols = c("#62B6CB", "#1B4965") # Setting the colours in the plot
)
```

Figure 2 here.

Findings from the 100 generated synthetic datasets indicate that 71% of datasets demonstrated the same inferential result (i.e., a statistically significant p -value). For the effect size, 57% of synthetic datasets maintained a ‘large’ effect size categorization.

Results for Studies 3 - 9

Studies in the domains of fluency, voice and resonance, communication modalities, receptive and expressive language, and social aspects of communication demonstrated more than 95% p -value agreement between the original and synthetic datasets (Table 3). Among studies that demonstrated lower agreement, the absolute mean difference between the synthetic p -values and the original p -value was 0.05 ($SD = 0.1$) for articulation, 0.03 ($SD = 0.04$) for hearing, and 0.25 ($SD = 0.28$) for cognitive aspects of communication (Figure 3). For effect size categorization agreement, studies in the domains of fluency, hearing, communication modalities, and cognitive aspects of communication maintained the effect size categorization of the original study. Among studies that demonstrated lower effect size categorization agreement, the absolute mean difference between the effect size from synthetic datasets and the original study’s effect

size was 0.19 ($SD = 0.12$) for articulation, 0.09 ($SD = 0.07$) for voice and resonance, 0.06 ($SD = 0.05$) for receptive and expressive language, and 0.21 ($SD = 0.2$) for social aspects of communication.

Figure 3 here.

Figure 4 here.

Table 4 here.

Discussion

Although computational reproducibility is a core principle of science, data sharing is uncommon in CSD, partly due to concerns regarding disclosure risk (Pfeiffer et al., 2024). This study demonstrates the utility of synthetic datasets to protect participant confidentiality while preserving the statistical properties and relationships of the original analysis data. The utility of synthetic data is further strengthened by the range of datasets included in the current study, which varied by domain (across nine ASHA domains), sample size (from 40 to >8,000 data points), statistical models (from simple correlations to multilevel model with 3-way interactions), and effect sizes (from conventionally “small” to “large”). These results suggest that synthetic datasets can be effectively used across a wide range of studies in the field of CSD to preserve participant confidentiality when sharing data.

One key finding is that lower agreement between synthetic and original datasets was not attributed to sample size, despite the *synthpop* package’s recommendation of a minimum of 130 observations for generating synthetic datasets (Nowok et al., 2016). For example, in the original study from the cognition domain, which included over 8,000 observations, only 35% of synthetic

datasets maintained the same inferential result as the original dataset. Instead, p -value and effect size agreement between the synthetic and original datasets was influenced by the original data's proximity to the statistical significance or effect size thresholds. For example, the original cognition study reported a p -value of .013, resulting in a 35% agreement rate for synthetic datasets. Conversely, studies that reported an original p -value of $<.001$ showed a p -value agreement rate of 97-100%, with the exception of the articulation study, which had a p -value agreement of 71%.

These findings highlight the importance of verifying the accuracy of synthetic datasets and providing these comparisons in supplemental manuscript materials. To ensure synthetic data quality, researchers should generate multiple versions of a synthetic dataset and select the one that most closely reproduces the statistical findings of the original analysis. If the synthetic dataset fails to sufficiently maintain these relationships, it should not be shared.

This study is not without limitations. We used predetermined thresholds (e.g., 'significant' p -values and effect size categories) to evaluate whether synthetic data maintained the relationships observed in the original study. When the original analyses had p -values near the threshold for significance (e.g., $.01 < p < .05$) or effect sizes near the boundary of a category, lower agreement was more likely. This likely reflects the distribution of synthetic data across both sides of these thresholds rather than actual poor agreement (Figures 3 & 4). Additionally, it's important to recognize that synthetic data is inherently a proxy and cannot entirely preserve all statistical properties of the original dataset. Therefore, researchers should provide de-identified (or identifiable when ethical approval is obtained) data whenever possible, as well as evaluate the utility of the synthetic dataset in the context of their own study. Finally, open data alone does not ensure computational reproducibility. Instead, both open data and accompanied

code or syntax is required to reproduce analyses. In fact, recent research showed that a high percentage of findings from registered reports that provided open data were unable to be reproduced (Obels et al., 2020). Reproducible workflows in languages like R have been proposed and warrant consideration (Peikert et al., 2021).

Data Sharing Framework

In this framework, we aim to empower researchers who may feel uncertain or unmotivated to consider data sharing for their current or future work. We begin by examining the scientific and ethical implications of closed data, followed by an evaluation of the commonly used “available upon request” approach to data sharing, which we argue is insufficient. Finally, we outline the benefits of open data practices, emphasizing that sharing different types of data (raw, intermediate, analysis, and synthetic) can offer various levels of utility and impact.

Ethical and Scientific Need for Open Data

Closed data impedes cumulative science and raises ethical concerns. Researchers have an ethical responsibility to maximize the use of clinical data, as participants typically enroll in research with the expectation that their data will help answer important public health questions. In fact, studies on participants’ motivations, particularly in non-experimental, observational research where there is no direct benefit, show that altruism is a key factor for participation (i.e., “I signed up because this study might be able to help future patients in my situation”) (Soule MD et al., 2016). Thus, it is essential that researchers ensure participants’ data is fully utilized to benefit future patients, clinical outcomes, and scientific knowledge.

Additionally, many research questions cannot be fully answered in a single study due to limitations like small or unrepresentative samples. Sharing data extends the value of collected

datasets and allows other researchers to build on previous findings. Closed data practices also place an undue burden on future participants, particularly in studies involving invasive methodologies (e.g., radiation from videofluoroscopic swallow studies or neuroimaging) or require extensive travel and time. If previously collected data is not shared, future participants may undergo unnecessary procedures to duplicate these data.

Researchers conducting publicly funded studies are also ethically obligated to return data to the public that financed their research, a responsibility increasingly emphasized by funding agencies such as the National Institutes of Health and the National Science Foundation (Watson et al., 2023; Wilkinson et al., 2016). Even in cases where research is privately funded, participants arguably have the right to see their data shared and used to its fullest potential. In our experience, when participants are informed during the consent process about the potential for their data to be shared and reused, they overwhelmingly support data sharing to maximize the impact of their contribution.

Although some researchers may consider data availability statements like “available upon reasonable request” as a step toward data sharing, recent research has shown poor compliance with less than half of studies providing requested data (Tedersoo et al., 2021). In many cases, researchers do not devote the time to properly organize their data, thereby hindering its availability when requested or may restrict access to protect their data from reuse. Moreover, purposefully vague and unclear data availability statements may exacerbate inequities in the field. This practice limits access, particularly for those with fewer resources or opportunities, and poses a direct barrier to a cumulative and transparent scientific literature.

Benefits of Open Data

Open data offers substantial benefits for both the scientific community and researcher.

For example, studies show that openly shared data is associated with higher citation rates for the original work (Drachen et al., 2016; Piwovar et al., 2007; Piwovar & Vision, 2013). However, sharing data is not always straightforward, and researchers must consider when and what types of data to share. Figure 5 proposes a decision tree to guide researchers through the process of deciding which type of data to share. For practical guides on obtaining consent, sharing data, and ensuring FAIR data principles, we direct our readers to (Ohmann et al., 2017). Different types of data can be shared, including raw, intermediate, analysis, and synthetic data (Table 1), each providing varying levels of utility and benefit.

Figure 5 here.

Analysis Data. Sharing analysis data enables others to verify results during peer review or post-publication, promoting a more transparent and reliable scientific record. This practice also allows science to be self-correcting (Vazire & Holcombe, 2022). Analysis data is particularly important for meta-analyses, as these depend on comprehensive reporting of descriptive statistics. Unfortunately, many studies do not report all necessary details (e.g., means, standard deviations, and sample sizes) or use different statistical analyses, making it challenging to synthesize results across studies. Sharing analysis datasets can fill this gap, allowing more studies to be included in meta-analyses and resulting in more robust analyses (e.g., individual participant meta-analysis) (Eisenhauer, 2021; Yu & Romero, 2024) and conclusions (Chow et al., 2023), which is especially valuable for studying low-incidence populations in fields like CSD.

Raw or Intermediate Data. Sharing raw or intermediate data further enhances transparency by enabling researchers to reproduce the calculations behind analysis data. Different operational definitions or analysis steps are often a barrier to inclusion in a meta-

analysis. Sharing this type of data ensures that secondary analyses can be performed with alternate methodologies or operational definitions, as the field progresses. In this sense, sharing raw or intermediate data facilitates the generation of new knowledge and accelerates scientific discovery. Despite its many benefits, there are instances where sharing raw or intermediate data may not be feasible. For example, researchers may not have obtained consent from participants for data sharing, or the institutional review board may impose project-specific guidelines that restrict sharing this type of data. Even de-identified data may carry risks, as participants could be re-identified through indirect identifiers.

Synthetic Data. In such cases, synthetic data offers a viable alternative. Synthetic data maintains the statistical properties of the original data while protecting participant privacy, thus facilitating computational reproducibility. In this study, we demonstrated how synthetic data can be generated using the *synthpop* package in R across a wide range of datasets in the field of CSD. Recognizing that coding expertise may be a barrier for some researchers, we have also developed a free Shiny website that interfaces with *synthpop*, allowing researchers to easily generate synthetic versions of their data (https://csdsynthetic.shinyapps.io/synthetic_data_generation/).

Moving Forward

Current training models and incentive structures are not well-equipped to promote data sharing. To encourage open science practices like data sharing, larger systemic changes are likely necessary at both the organizational level (e.g., ASHA, societies) and within academic institutions. For example, doctoral programs should offer coursework that introduces these concepts and educates future researchers on best practices for data sharing. Fortunately, many resources are available for current researchers to familiarize themselves with these practices

(Lewis, 2024). Additionally, institutions must incentivize data sharing and recognize open science efforts as valuable scholarly contributions. Although ASHA has introduced open science badges to acknowledge these efforts, it remains unclear if this is enough to encourage large-scale participation. Ultimately, a broader cultural shift is needed in the field - from the current individualistic, siloed approach to a more collaborative and pro-social view of science.

Conclusions

This study assessed the utility of the *synthpop* package in R for generating synthetic data in situations where sharing original data poses risks of participant re-identification. We demonstrated that synthetic data can be effectively applied across various data types and research areas within the CSD field. In most cases, the synthetic data closely matched the *p*-values and effect sizes of the original data, though a few instances showed lower agreement. Therefore, researchers using synthetic data should verify its accuracy in reproducing their original findings before sharing. Finally, we provide a framework for data sharing, emphasizing that whether researchers share raw, intermediate, analysis, or synthetic data, some form of data sharing is achievable for all. Our overarching goal is to establish data sharing as the standard practice, rather than the exception, in CSD.

487 **Acknowledgements**

488 We would like to thank the authors of the studies included in this manuscript for making
489 their data publicly available.

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Table and Figure Captions

Table 1: Description of types of data.

Table 2: Characteristics of included studies by ASHA domain.

Table 3: Effect size measures and interpretation by statistical test.

Table 4: Stability of synthetic datasets across ASHA domains.

Figure 1. Visualization of data distributions from synthetic and original data for Study #1 (Curtis et al., 2023).

Caption: Panel A displays the overall distribution of laryngeal vestibule residue. Panel B displays the frequency of values by bolus consistency.

Figure 2. Visualization of data distributions from synthetic and original data for Study #2 (Thompson et al., 2023).

Caption: Panel A displays the distribution of vowel space area and panel B displays the distribution of speech intelligibility.

Figure 3. Distribution of log-transformed p -values in synthetic datasets across ASHA domains.

Caption: Each panel displays the distribution of log-transformed p -values across 100 synthetic datasets for a given ASHA domain. The dashed line indicates the threshold for statistical significance from the original study. Shaded green areas indicate synthetic p -values that

maintained the statistical inferential result of the original study. The mean difference and standard deviation of raw p -values compared to the p -value reported in the original study is shown below each panel's title.

Figure 4. Distribution of effect sizes in synthetic datasets across ASHA domains.

Caption: Each panel displays the distribution of effect sizes across 100 synthetic datasets for a given ASHA domain. The dashed line indicates the effect size reported in the original study and the light blue shaded area indicates the range of the effect size categorization. The mean difference and standard deviation of the effect size compared to the result reported in the original study is shown below each panel's title.

Figure 5. Decision tree for data sharing.

Figure 1

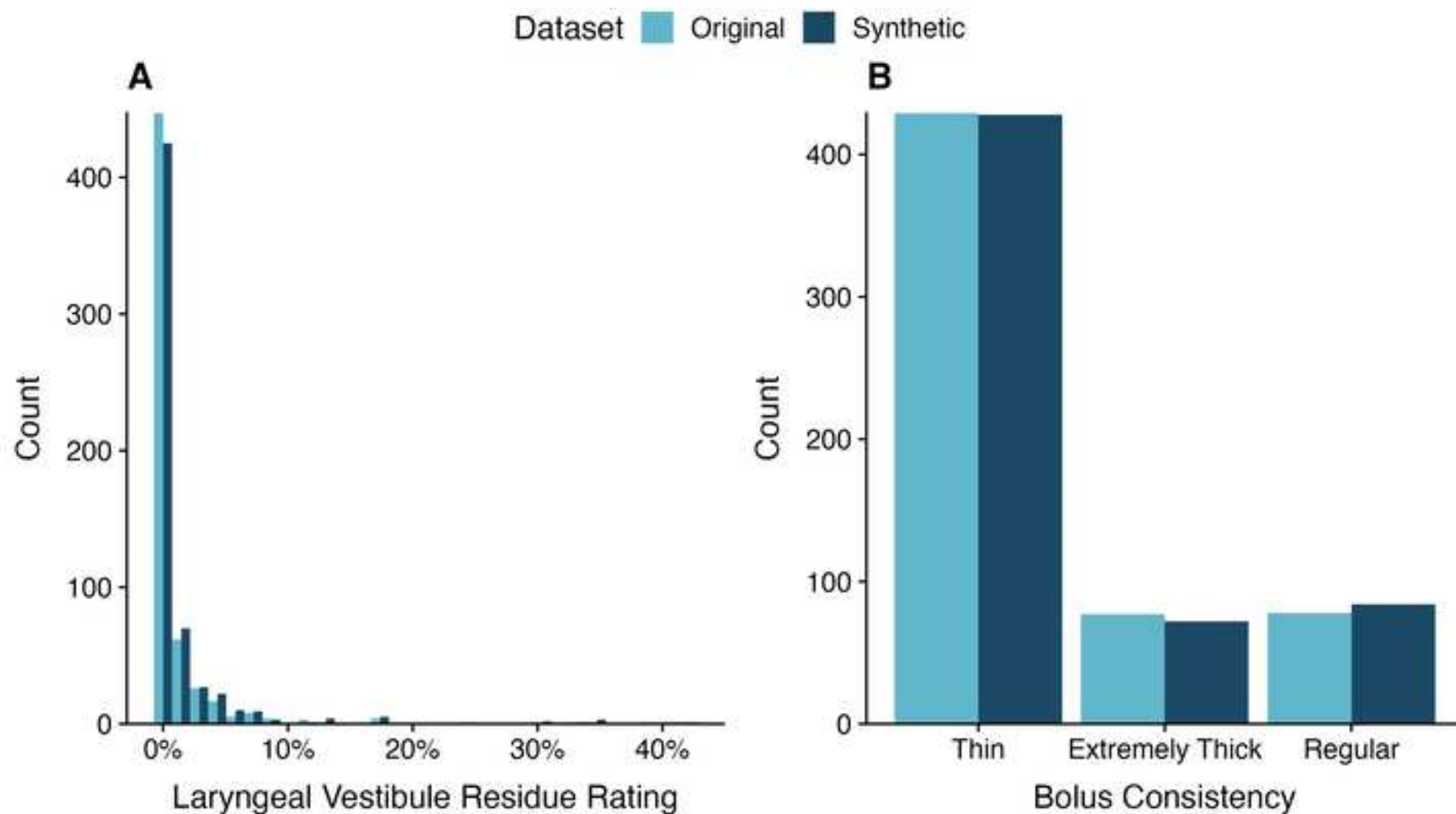
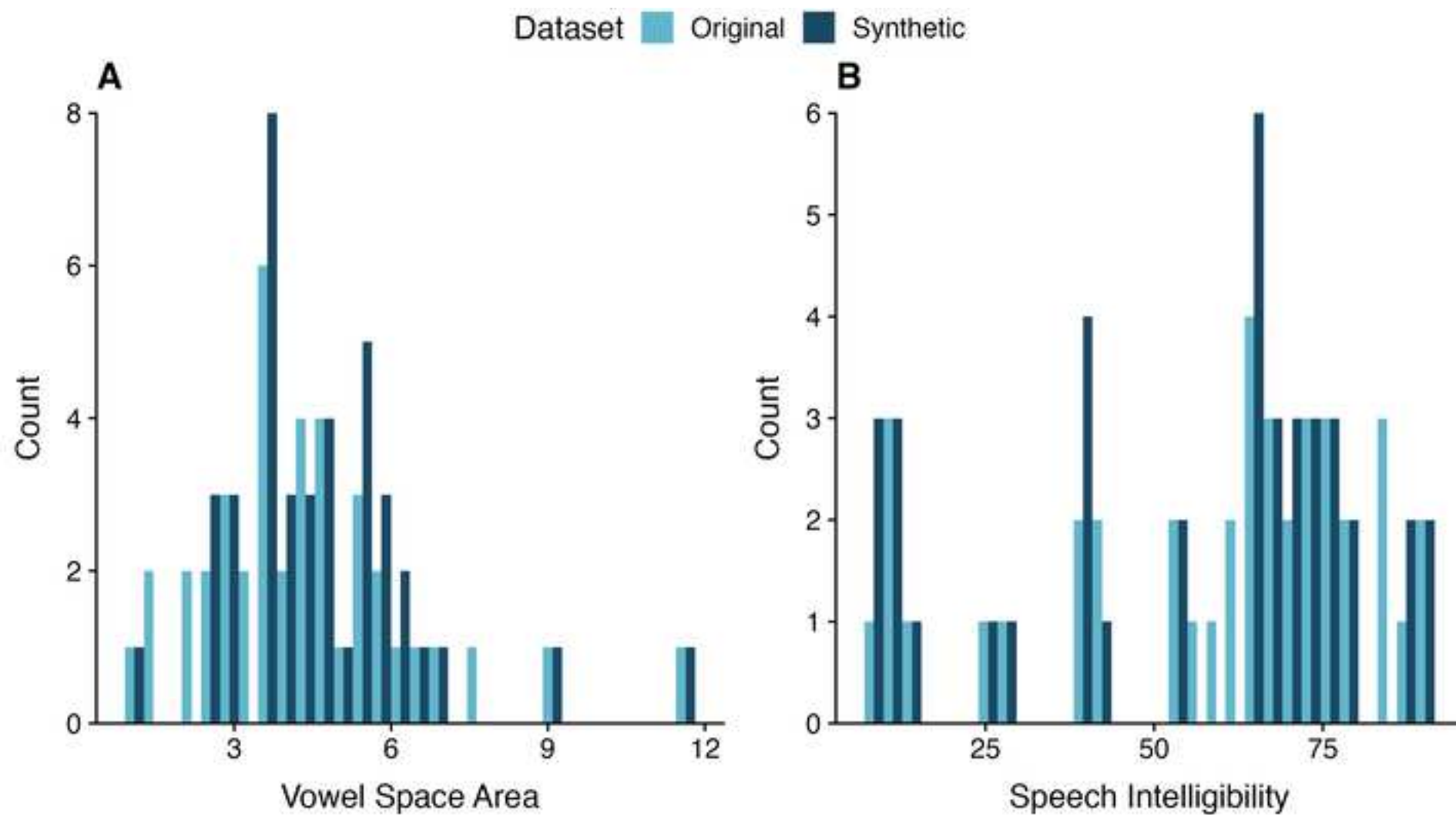
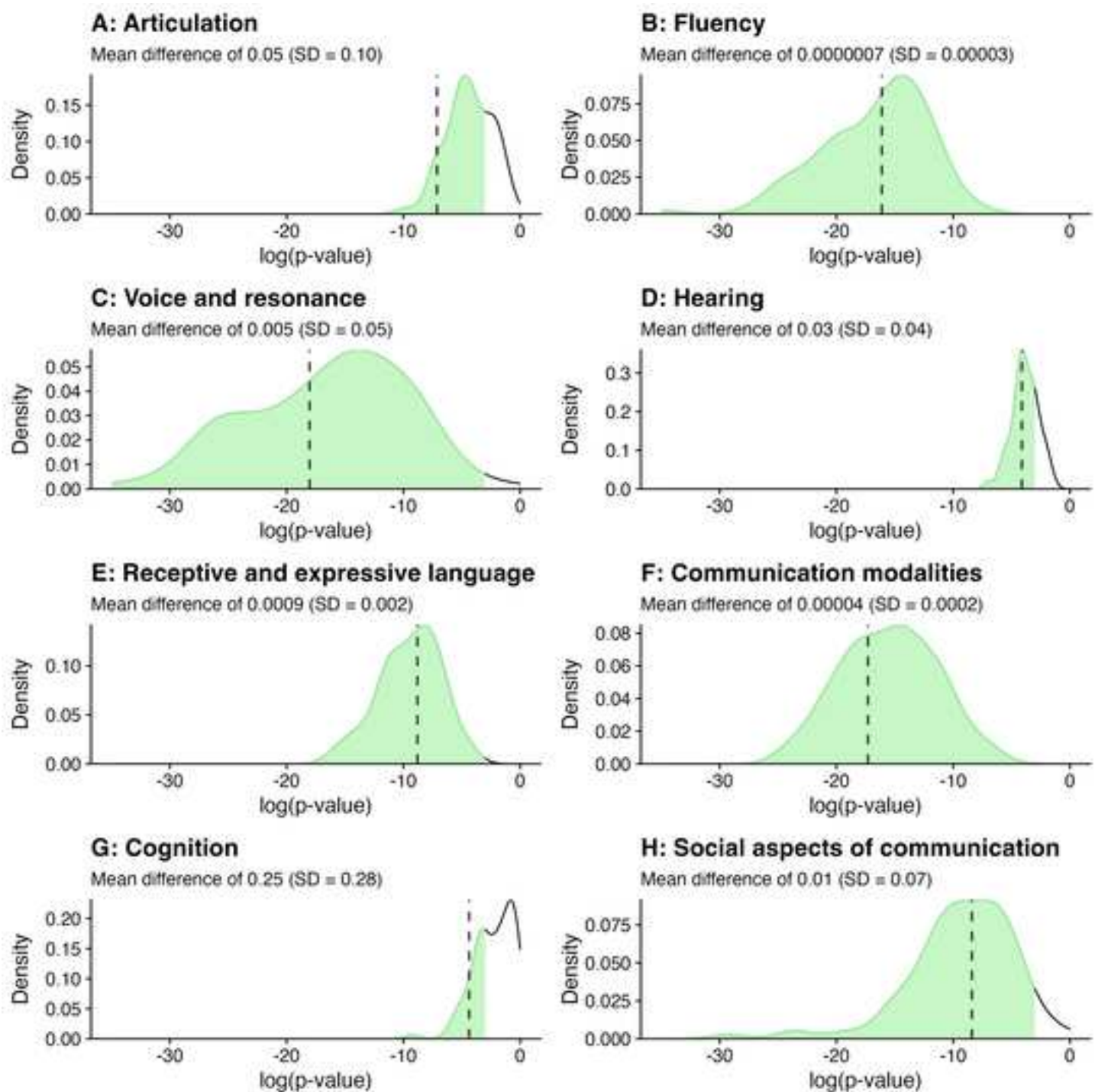
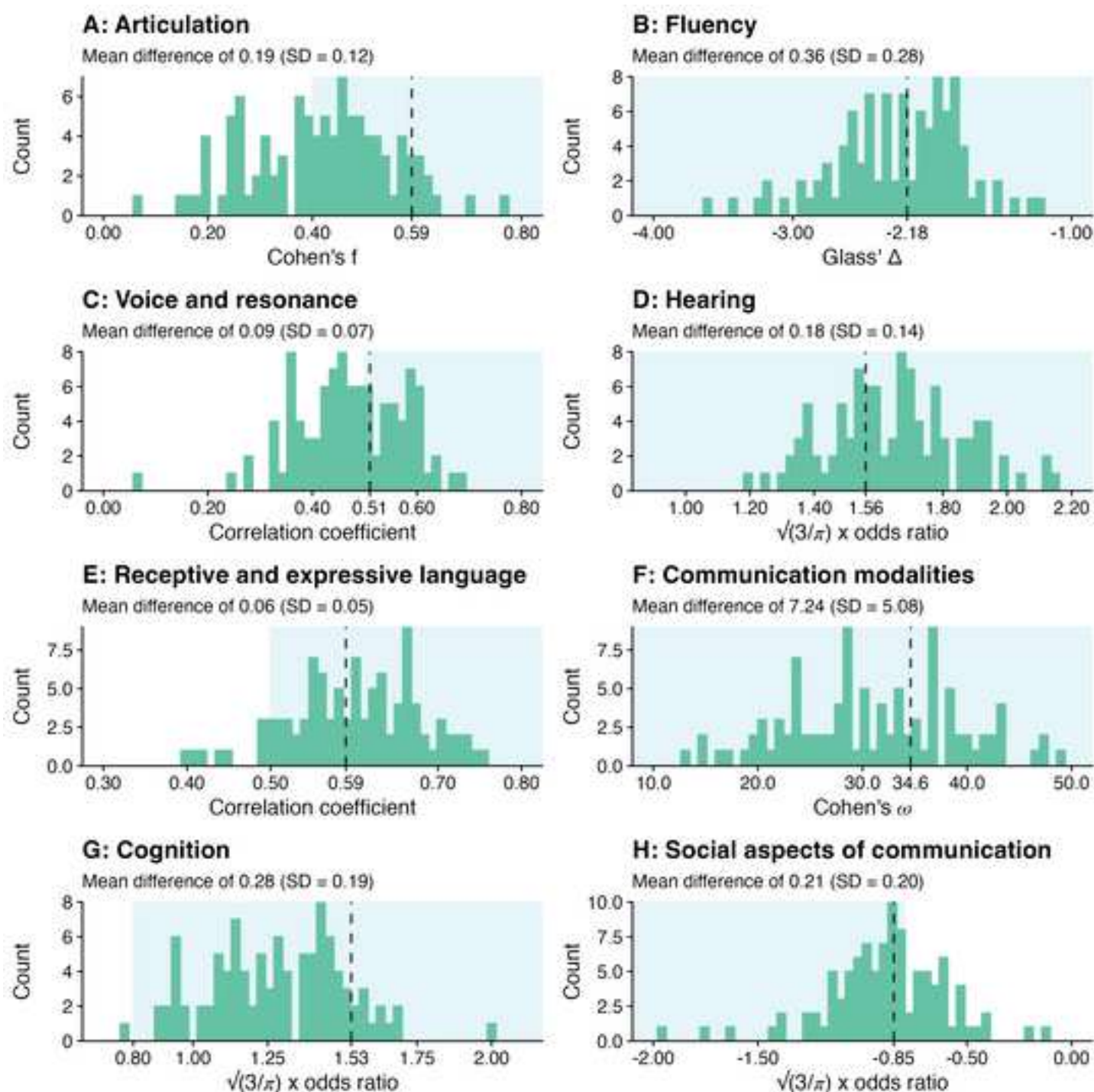


Figure 2

[Click here to access/download;Figure;figure_2.png](#)







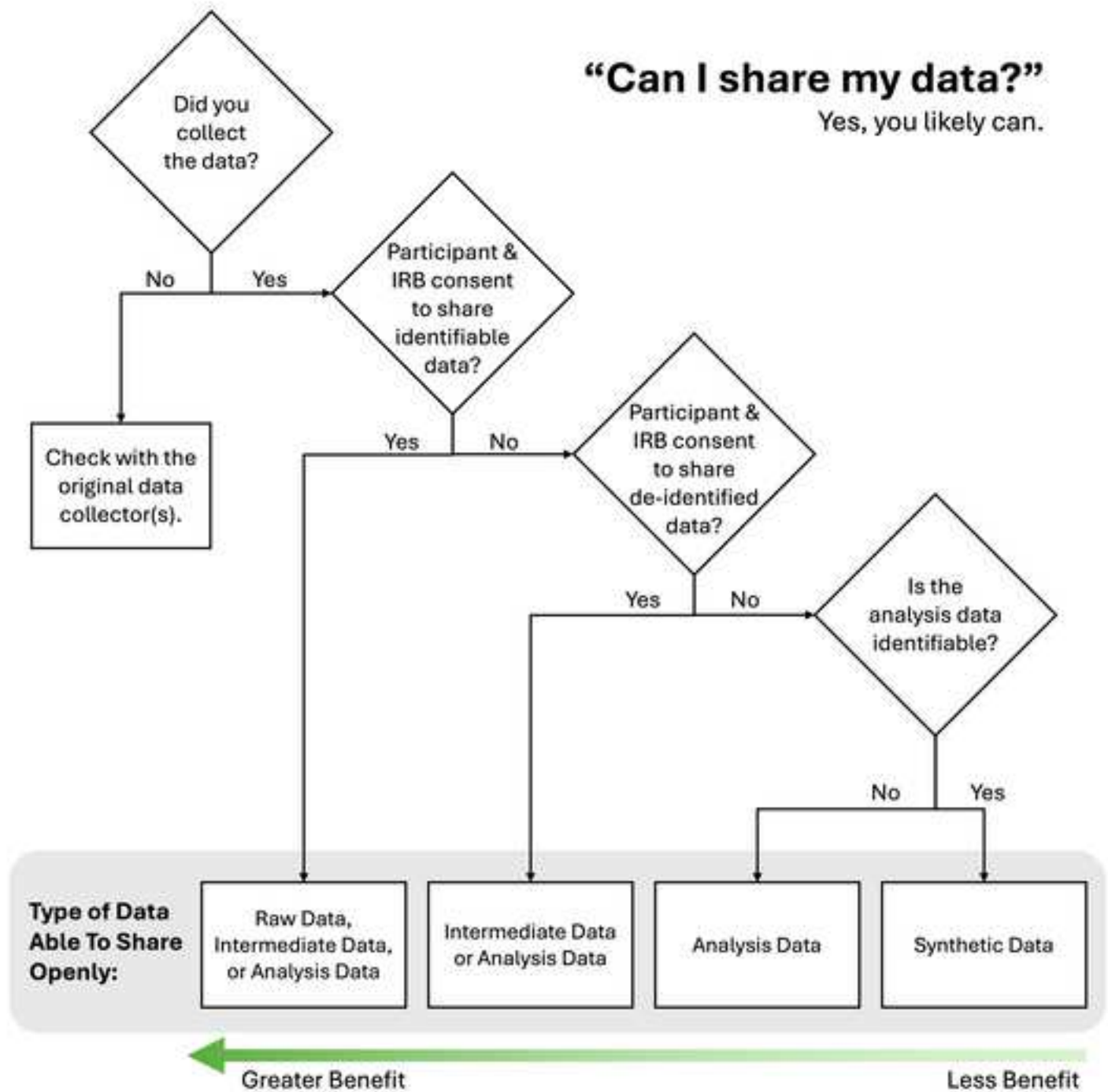


Table 1: Description of types of data.

	Raw Data	Intermediate Data	Analysis Data
Description	Original, unmodified data collected from studies. Also known as primary data, microdata, individual-level data.	Cleaned, de-identified, and processed data, used for creating the analysis data. Also known as cleaned data, transactional data, processed data.	Final dataset containing variables used in statistical analyses. Also known as derived data, result data, aggregate data.
Examples			
Acoustic Data (Thompson et al., 2023)	Raw audio recordings.	Extracted formant data.	Dataset containing average vowel space area per speaker.
Swallowing Data (Curtis et al., 2023)	Video files from flexible endoscopic evaluations of swallowing.	Not applicable.	Dataset containing ratings of swallowing safety and efficiency.
Eye Tracking Data (Baron et al., 2023)	Eye movement recordings (gaze, saccades, fixations).	Cleaned data with merged fixations and removed artifacts.	Dataset containing summary of fixation durations, reading times, and target proportions.
Focus Group or Interview Data (Pfeiffer et al., 2024)	Audio recordings and transcripts.	De-identified transcripts.	Dataset containing coded categories and themes from qualitative analysis.
Survey Data (Riccardi, 2024)	Raw survey responses.	Cleaned and coded responses.	Dataset containing summary scores and frequencies of survey responses.
Assessment Tool Data (Pfeiffer & Landa, 2024)	Raw, unscored assessment protocols.	Scored protocols with calculated totals and subscales.	Dataset containing standard scores, confidence intervals, percentile ranks.

Table 2: Characteristics of included studies by ASHA domain.

Domain	Study	Open Materials	Sample Size	Population(s)	Analysis of Interest	Outcome Type(s)	Statistics
Swallowing	Curtis et al. (2023)	Data, code	39	Neurotypical	Distribution of laryngeal vestibule residue ratings	Continuous	Descriptive
Articulation	Thompson et al. (2023)	Data, code	40	Parkinson's disease, amyotrophic lateral sclerosis, Huntington's disease, cerebellar ataxia	Relationship between vowel space area and intelligibility	Continuous	Hierarchical linear regression
Fluency	Elsherif et al. (2021)	Data, code	164	Dyslexia, stuttering, neurotypical	Group difference in nonword repetition	Continuous	Independent t-test
Voice and resonance	Novotný et al. (2016)	Data	111	Parkinson's disease, Huntington's disease, neurotypical	Relationship between overall perceptual rating and variability of nasality	Continuous	Pearson correlation
Hearing	Battal et al. (2019)	Data, code	34	Congenitally blind, sighted	Group difference in auditory localization	Continuous	Linear mixed-effects model with 3-way interaction
Communication modalities	King et al. (2022)	Data, code	160	Speech-language pathologists	Timepoint difference in lack of/limited internet and technology barriers	Ordinal	Chi-square

Domain	Study	Open Material s	Sampl e Size	Population(s)	Analysis of Interest	Outcome Type(s)	Statistics
Receptive and expressive language	Kearney et al. (2023)	Data	34	Brain tumor	Relationship between years of education and reading score	Continuous	Spearman's rank correlation coefficient
Cognitive aspects of communication	Clough et al. (2023)	Data	102	Traumatic brain injury, neurotypical	Group x Condition interaction in emotion recognition accuracy	Binary	Generalized linear mixed-effects model with 3-way interaction
Social aspects of communication	Chanchaochai & Schwarz (2023)	Data, code	96	Autism spectrum disorder, neurotypical	Group difference in non-verbal IQ	Continuous	Analysis of Variance

Table 3: Effect size measures and interpretation by statistical test.

Statistical Test	Effect Size Measure	Small	Medium	Large
Hierarchical linear regression	Cohen's f (Cohen, 1988)	0.1	0.25	0.4
Independent t-test	Cohen's d (Cohen, 1988) & Glass' Δ (Hedges & Olkin, 1985)	0.2	0.5	0.8
Correlation (Pearson's, Spearman's)	Correlation coefficient (Cohen, 1988)	0.1	0.3	0.5
Chi-square	Cohen's ϕ (Cohen, 1988)	0.1	0.3	0.5
Generalized linear model	$\sqrt{(3/\phi^2)} \times \text{odds ratio}$ (Haddock et al., 1998; Hasselblad & Hedges, 1995)	0.2	0.5	0.8

Table 4: Stability of synthetic datasets across ASHA domains.

Domain	Study	Sample Size	P-value ^a	Effect Size Measure	Effect Size	P-value Agreement	Effect Size Categorization Agreement
Swallowing	Curtis et al. (2023)	584	<.001	$\sqrt{(3/\square)} \times$ odds ratio (μ)	.06	100% (zero-inflated) & 98% (beta)	100% (zero-inflated) & 100% (beta)
Articulation	Thompson et al. (2023)	40	<.001	Cohen's f	.59	71%	57%
Fluency	Elsherif et al. (2021)	114	<.001	Glass' Δ	-2.18	100%	100%
Voice and resonance	Novotný et al. (2016)	111	<.001	Correlation coefficient	.51	99%	40%
Hearing	Battal et al. (2019)	372	.016	$\sqrt{(3/\square)} \times$ odds ratio	1.56	78%	100%
Communication modalities	King et al. (2022)	2,160	<.001	Cohen's $\square$	34.6	100%	100%
Receptive and expressive language	Kearney et al. (2023)	36	<.001	Correlation coefficient	.59	100%	90%
Cognitive aspects of communication	Clough et al. (2023)	8,568	.013	$\sqrt{(3/\square)} \times$ odds ratio	1.53	35%	99%
Social aspects of communication	Chanchaochai & Schwarz (2023)	96	<.001	Cohen's d	-.85	97%	66%

^aCaption: 100 synthetic datasets were generated for each domain. For the Swallowing domain, a zero-inflated beta multilevel model was performed to directly compare original and synthetic datasets. The percentage of non-significant p-values, which is indicative of no statistically significant difference between dataset types, is shown for both zero-inflated and beta portions of the model.